MODULE *misbehavior*

Version 2 *misbehavior* and banning specification.

Models the receiving side of the draft's "Misbehavior and Banning" section: a local node scoring the peers it is connected to. The draft constrains every penalty by a provability principle — a node MUST NOT penalize behaviour that a conformant peer acting on honest but divergent state could produce — and motivates it: penalties on weaker evidence let an attacker induce honest nodes to ban one another. This module turns that sentence and its converse into checkable properties.

Remote peers are honest or *Byzantine*. Honest peers may legitimately send:

divergent headers	contiguous, valid proof of work, but not connecting to the local chain: the peer follows another fork ("Block Announcements", "Headers-First Synchronization" — MUST NOT penalize).
requested invalid	a block the local node itself requested by hash whose content fails consensus validation: the responder served exactly the bytes the hash names; blame lies with the announcer ("Misbehavior and Banning", exemption "Content of requested objects").

Byzantine peers additionally send provable violations from the penalty table: oversize get-headers responses (+ 20), non-contiguous headers (+ 20), and an announced block with invalid proof of work (+ 100).

Three booleans select wrong readings of the draft:

$PenalizeNonConnecting = \text{TRUE} \rightarrow$	divergent headers score 20 points
$PenalizeRequestedInvalid = \text{TRUE} \rightarrow$	*a* requested invalid block scores 100
$PerConnectionScore = \text{TRUE} \rightarrow$	the score is per connection, reset on reconnect, instead of keyed by address

Under the first two, *NoHonestBan* is violated: an honest peer on another fork is banned. Under the third, *PersistentAttackerBanned* is violated: a *Byzantine* peer sheds its score by reconnecting before the threshold — the draft's stated reason for address-keyed persistent scores.

See ../documents/v2-*modeling.md* for the full write-up.

EXTENDS *Naturals*, *FiniteSets*

CONSTANT *Peers*	remote peers of the local node
CONSTANT *Byzantine*	subset of *Peers* sending provable violations
CONSTANT *BanThreshold*	score at which a peer is disconnected and banned
CONSTANT *PenalizeNonConnecting*	TRUE penalizes divergent headers (buggy)
CONSTANT *PenalizeRequestedInvalid*	TRUE penalizes requested invalid blocks (buggy)
CONSTANT *PerConnectionScore*	TRUE resets the score on reconnect (buggy)

$Honest \triangleq Peers \setminus Byzantine$

VARIABLES

score,	$[Peers \rightarrow 0 \,..\, BanThreshold]$ *misbehavior* score, capped
banned,	$[Peers \rightarrow \text{BOOLEAN}]$ banned addresses (absorbing)
connected	$[Peers \rightarrow \text{BOOLEAN}]$ currently connected

$vars \triangleq \langle score, banned, connected \rangle$

$Min(a, b) \triangleq \text{IF } a < b \text{ THEN } a \text{ ELSE } b$

$Init \triangleq$

$\quad \land\ score \quad\ \ = [p \in Peers \mapsto 0]$

$\quad \land\ banned \quad = [p \in Peers \mapsto \text{FALSE}]$

$\quad \land\ connected = [p \in Peers \mapsto \text{TRUE}]$

The local node adds *pts* to *p*'s score; at the threshold it closes the connection with *MISBEHAVIOR* and bans the address ("Misbehavior and Banning": score thresholds and banning).

$Penalize(p, pts) \triangleq$

$\quad \text{LET } s \triangleq Min(score[p] + pts, BanThreshold)$

$\quad \text{IN } \quad \land\ score' \quad\ \ = [score \text{ EXCEPT } ![p] = s]$

$\qquad\qquad \land\ banned' \quad = [banned \text{ EXCEPT } ![p] = s \geq BanThreshold]$

$\qquad\qquad \land\ connected' = [connected \text{ EXCEPT } ![p] = @ \land s < BanThreshold]$

$NoPenalty \triangleq \text{UNCHANGED } \langle score, banned, connected \rangle$

An honest peer on a divergent fork answers get-headers, or announces a header, that does not connect to the local chain. The draft forbids a penalty; *PenalizeNonConnecting* scores it anyway.

$HonestDivergentHeaders \triangleq$

$\quad \exists\, p \in Honest :$

$\qquad \land\ connected[p]$

$\qquad \land\ \text{IF } PenalizeNonConnecting \text{ THEN } Penalize(p, 20) \text{ ELSE } NoPenalty$

An honest peer serves a block the local node requested by hash whose content fails consensus validation — exactly the bytes the hash names (for example an artifact held for checkpointed sync). The draft exempts it; *PenalizeRequestedInvalid* scores it anyway.

$HonestRequestedInvalid \triangleq$

$\quad \exists\, p \in Honest :$

$\qquad \land\ connected[p]$

$\qquad \land\ \text{IF } PenalizeRequestedInvalid \text{ THEN } Penalize(p, 100) \text{ ELSE } NoPenalty$

Provable violations from the penalty table: only *Byzantine* peers produce them, and they always score.

$ByzOversizeHeaders \triangleq$

$\quad \exists\, p \in Byzantine : connected[p] \land Penalize(p, 20)$

$ByzNonContiguousHeaders \triangleq$

$\quad \exists\, p \in Byzantine : connected[p] \land Penalize(p, 20)$

$ByzInvalidPowBlock \triangleq$

$\quad \exists\, p \in Byzantine : connected[p] \land Penalize(p, 100)$

A peer may disconnect at will — in particular a *Byzantine* peer trying to shed its score.

$Disconnect \triangleq$
$\quad \exists\, p \in Peers :$
$\quad\quad \land\ connected[p]$
$\quad\quad \land\ connected' = [connected\ \text{EXCEPT}\ ![p] = \text{FALSE}]$
$\quad\quad \land\ \text{UNCHANGED}\ \langle score,\ banned \rangle$

A disconnected, unbanned peer of S reconnects. Whether its score survives the new connection is the *PerConnectionScore* switch; the draft wants scores keyed by address precisely so that reconnecting sheds nothing.

$ReconnectFrom(S) \triangleq$
$\quad \exists\, p \in S :$
$\quad\quad \land\ \neg connected[p]$
$\quad\quad \land\ \neg banned[p]$
$\quad\quad \land\ connected' = [connected\ \text{EXCEPT}\ ![p] = \text{TRUE}]$
$\quad\quad \land\ score' = [score\ \text{EXCEPT}\ ![p] = \text{IF}\ PerConnectionScore\ \text{THEN}\ 0\ \text{ELSE}\ @]$
$\quad\quad \land\ \text{UNCHANGED}\ \langle banned \rangle$

$Reconnect \triangleq ReconnectFrom(Peers)$

$Next \triangleq$
$\quad \lor\ HonestDivergentHeaders$
$\quad \lor\ HonestRequestedInvalid$
$\quad \lor\ ByzOversizeHeaders$
$\quad \lor\ ByzNonContiguousHeaders$
$\quad \lor\ ByzInvalidPowBlock$
$\quad \lor\ Disconnect$
$\quad \lor\ Reconnect$

Fairness encodes the hypothesis of *PersistentAttackerBanned*: a *Byzantine* peer that KEEPS misbehaving — and keeps reconnecting when disconnected — is the one that must end up banned. Fairness is on the *Byzantine* peer's own reconnection (an honest peer reconnecting must not discharge it); honest events and *Disconnect* are never forced. The configurations use a single *Byzantine* peer, so the existential fairness cannot starve one attacker in favour of another.

$Spec \triangleq$
$\quad Init$
$\quad \land\ \Box[Next]_{vars}$
$\quad \land\ \mathrm{SF}_{vars}(ByzOversizeHeaders)$
$\quad \land\ \mathrm{SF}_{vars}(ReconnectFrom(Byzantine))$

Liveness: a persistently misbehaving *Byzantine* peer is eventually banned. Violated under *PerConnectionScore*: reconnecting before the threshold sheds the score forever.

$PersistentAttackerBanned \triangleq \Diamond \forall\, p \in Byzantine : banned[p]$

Safety invariants.

$$
\begin{aligned}
TypeOK \triangleq\ & \\
& \land\ score \in [Peers \rightarrow 0 \mathrel{..} BanThreshold] \\
& \land\ banned \in [Peers \rightarrow \text{BOOLEAN}] \\
& \land\ connected \in [Peers \rightarrow \text{BOOLEAN}]
\end{aligned}
$$

The provability principle, as an outcome: no honest peer is ever banned, however divergent its chain view. Violated under either buggy penalty.

$$
\begin{aligned}
NoHonestBan \triangleq\ & \\
& \forall\, p \in Honest : \neg banned[p] \land score[p] < BanThreshold
\end{aligned}
$$

A banned peer is disconnected and stays banned.

$$
\begin{aligned}
BanIsFinal \triangleq\ & \\
& \forall\, p \in Peers : banned[p] \Rightarrow \neg connected[p]
\end{aligned}
$$
